(c) Soil and Irrigation Parameters

It has been said that you don't irrigate a crop, you irrigate the soil. In order to understand the principles of irrigation, it is imperative that a person who works with irrigation keep the following in mind:

1. Water is first introduced to the soil surface and then to the soil profile either by snowmelt, rainfall or irrigation.
2. The crop then removes the water from the soil.

When water is to be introduced to the soil, there are two things that an irrigator is interested in:

1. How fast can the soil take the water?
2. How much water can the soil profile hold?

Intake rates (inches/hr) are used to describe or measure how fast the soil can take water. Intake rates are described in three ways:

*1. Initial Intake Rate is the rate the soil will take water when it is dry and water first contacts the surface.

*2. Basic Intake Rate is the rate the soil will take water after it is saturated.

*3. Average Intake Rate is the average rate of intake of a soil from a dry to a saturated condition.

*In these cases, this is without runoff.

Curves have been developed which show the intake rate of different soils plotted against time. The soils have been grouped together in "Intake Families."

An Intake Family of soils has the same average intake rate. Figure 2-3 (pg 2-18) & Figure 2-4 (pg. 2-19) is a plotting of the Family Intake Curves the NRCS uses.

To understand how much water can or will be introduced to a soil profile, the irrigator and technician need to become familiar with the following terms.

Wilting Point - The amount of moisture a soil profile will have when the plant is unable to obtain sufficient moisture to continue active growth.

Permanent Wilting Point - The amount of moisture a soil profile will have when the plant is unable to obtain any more moisture from the soil.

Saturation Point - The maximum amount of moisture a soil profile can hold.

Available Water Capacity (AWC) - How much water is available to the plant, the difference between the Field Capacity and the Permanent Wilting Point.

Figure 2-1 (pg 2-9) shows a graphical example of these parameters for different soil textures.

When we talk about the amount of moisture in the soil profile, it becomes evident that we are considering one specific depth.

The crop that is being grown and the stage of development it is in will dictate what depth of profile should be considered. See Nebraska supplements to Chapter 3 for root depths by crop and growth stage. Normal irrigation depths can be found in Nebraska supplements to Chapter 6, by crop, irrigation group, and slope group. It was stated earlier that the Available Water Capacity (AWC) is how much water is available to the plant. Although this water is available to the plant, in most cases the roots can't extract the moisture fast enough to keep the plant from stressing as it approaches the Permanent Wilting Point, see Figure 2-1 (pg 2-9).

If it is the irrigator's intention to irrigate before the plant is stressed, irrigation should begin before the Wilting Point. In this Irrigation Guide, we have said this will be when half of the water is used between the Field Capacity and the Permanent Wilting Point or half of the Available Water Capacity (AWC). This amount of moisture is called Management Allowable Depletion (MAD).

One half of the AWC was selected as an average for all soils. In reality, it will vary with different soils, crops, and climate. See “Consumptive Use of Selected Nebraska Crops” table in Nebraska Supplements to Chapter 4.

(d) Use of Irrigation Design Groups

The soils listed in this Irrigation Guide include the irrigable soils presently being mapped in Nebraska. Land types and soils genera1ly considered non-irrigable are not included. As additional soils are recognized, they are to be added to the appropriate irrigation design group. Some soil phases or variants suitable for irrigation are not included in the listing of soils. The design group will be assigned locally in counties where they occur.

The soils in each series were evaluated and placed into one of 14 groups called Irrigation Design Groups. Soils having approximately equal intake rates, available water capacities, and available root zone depths were placed together. Some groups include soils with minor variations in intake rate, available water capacity and permeability. If the variation is significant, it is noted in the irrigation design group where the soil is listed.

The grouping of soils is shown in the two listings that follow. The first list shows all the soils in alphabetical order by series, surface texture with the appropriate irrigation design group.

The second listing is by irrigation design groups and gives the principal soils included in each of the 14 groups. Two of these are divided into sub- groups because of minor profile differences and avai1able water capacity. This is not enough difference to justify dividing the soils into additional design groups. The intake family used in preparing the design data is included.

Included is a general description of the texture profiles. Where soils of a design group are subdivided, there is a general description for each subgroup.

The estimated available water capacity for each soil group or subgroup follows the description. The amounts of moisture are cumulative by one-foot or one-half foot increments of depth. The data used to calculate the available water capacities are in Table 2-16 that follows this grouping of soils. This table shows available water capacities by soil texture classes. It also gives the range of availab1e water capacity for each of the soil texture classes for the surface layer, subsoil, and lower horizon. With this table it is possible to make a reasonable estimate of the available water capacity of any soil profile for which the horizon thickness and texture is mown.

The most common soils and their respective field symbol are listed below the available water capacities for each design group or subgroup.

A form for listing each soil and the irrigation design group number in a county or survey area is included last in this section. Listing the mapping unit or soils in a county on one or several pages of this form simplifies the use of this guide for each area. The statewide list that follows this section will be of value in preparing the county list.

Available Water Capacity of Soils by Soil Texture Classes

In estimating available water capacity of the groups of soils in this irrigation guide, the predominant soil textures of the major soil horizons for soils in each irrigation group were used. Individual soils in the irrigation design groups vary somewhat in total available water capacity because of minor differences in texture of the soil horizons.